E-AQI: Edge-Aware Quality Index for Perceptual Clipping Assessment in Enhanced Images

Ali Hosseini[a,*], Saeed Sedighi[a]

[a]*Department of Computer Engineering, Shiraz Branch, Islamic Azad University, Shiraz, Iran*

Abstract

Clipping assessment requires a distinction between code-value saturation and perceptually objectionable loss of information. Saturation near a structural boundary may have a different effect on quality models from saturation that removes interior texture, but the direction and magnitude of this difference require controlled measurement. We formulate E-AQI, a no-reference clipping-severity framework that combines a two-sided adaptive saturation map, structural-edge evidence, and saliency and local entropy features. The formulation separates highlight and shadow thresholds, preserves numerical behavior under bit-depth rescaling, and provides a smooth histogram surrogate for a potential enhancement-training loss. Three lines of evidence are reported. First, two computational studies use established no-reference quality measures as measurement instruments on controlled stimuli: on 168 area-matched edge-aligned versus interior-texture clipping pairs, classical measures (BRISQUE, NIQE, PIQE) show no significant sensitivity to clipping location (two-sided Wilcoxon signed-rank $p \geq 0.066$), whereas a deep measure (TOPIQ-NR) and the trained E-AQI prototype rate interior-texture clipping as worse in 71–72% of pairs ($p = 2.4 \times 10^{-8}$ and $p = 2.3 \times 10^{-11}$ respectively); and across 480 enhancement levels on 60 images, the Spearman rank-order correlation coefficient (SRCC) between the clipped-area fraction and every learned measure stays below $|\rho| = 0.30$. Second, the trained E-AQI prototype and a non-edge clipped-area fraction descriptor (PCoE) are evaluated on the same controlled clipping stimuli, and the prototype (0.43M parameters; 21 ms at 256×192 and 227 ms at 512×384 on a consumer GeForce GTX 1050 Ti) is trained end-to-end on KonIQ-10k under a source-disjoint protocol, reaching SRCC $= 0.708 \pm 0.011$ and Pearson linear correlation coefficient PLCC $= 0.751 \pm 0.012$ (mean ± SD over three seeds), above all non-retrained baselines evaluated identically (best: PCoE, SRCC = 0.191) and below the much larger pretrained TOPIQ-NR model (0.879). Third, the preregistered paired human perceptual study on the released open-source platform is executed with 51 observers (126 base images, five matched conditions, 0–100 annoyance scale): interior-texture clipping is rated more annoying than edge-aligned clipping (mean paired difference +5.3 on the 0–100 scale, Wilcoxon $p = 3.4 \times 10^{-7}$, Cohen's $d_z = 0.89$, 82% of observers), and annoyance separates all five conditions (Friedman $\chi^2(4) = 191.8$, $p = 2.1 \times 10^{-40}$) with the ordering T > E > X > N > C, replicating the deep measure's location sensitivity perceptually. Reproducible synthetic checks verify operator consistency.

Keywords: No-reference image quality assessment, clipping, saturation, structural edges, low-light image enhancement, differentiable quality assessment

1. Introduction

Image enhancement can map distinct scene intensities to the same output code value. This saturation can remove texture or color information, yet saturated regions also occur in ordinary image content such as bright backgrounds and deep shadows. A useful assessor should therefore distinguish a saturation response from evidence that the response is objectionable. E-AQI investigates whether explicit conditioning on structural boundaries and local context improves that distinction.

BRISQUE [1], NIQE [2], and PIQE [3] assess image quality through statistical or local distortion descriptors. Contemporary approaches, including TOPIQ [4] and LIQE [5], incorporate content and spatial information. Their architectures do not justify the blanket claim that all existing metrics penalize every saturated pixel identically. The narrower question addressed here is whether an explicit clipping-and-boundary representation adds measurable value under controlled saturation changes.

While low-light recognition datasets such as ExDark [6] provide authentic underexposed scenes, they do not include pristine reference counterparts or human perceptual quality ratings; measuring perceptual clipping therefore requires controlled stimuli or dedicated enhancement-quality benchmarks with human opinion scores. This observation motivates the paired, area-matched design used throughout this work.

This paper makes four contributions. First, a consistent two-sided clipping formulation with synthetic numerical checks that are reproducible without model training; all values reported in Section 4 are produced by the accompanying reference implementation. Second, two controlled computational studies (Section 5) that use established no-reference quality measures as measurement instruments: (i) an area-matched

*Corresponding author.
Email address: ali.hosseini7148@iau.ac.ir (Ali Hosseini)

clipping-location experiment on 168 pairs built from real photographs, and (ii) a clipping-dose response experiment across 480 enhancement levels on 60 images. These studies quantify how strongly existing measures respond to where—and how much—clipping occurs, and Section 5.1 extends the same stimuli to the trained E-AQI prototype and to the PCoE clipping descriptor. Third, a profiled prototype implementation (Section 5.3) trained end-to-end on KonIQ-10k under a source-disjoint protocol, together with an executed component-wise ablation (Section 6.7) that isolates the clipping, edge, and context branches. Fourth, a preregistered paired human perceptual study (Section 5.4), executed on the released open-source web platform with 51 observers: the stimulus pipeline, consent and screening flow, rating interface, and the fixed analysis plan are released together with the collected ratings so that the perceptual validation can be independently replicated.

2. Related Work and Scope

2.1. General No-Reference Quality Assessment

Statistical models such as BRISQUE [1] and NIQE [2] remain useful controls because they are computationally accessible and do not explicitly encode a clipping-edge interaction. PIQE [3] is a third cheap classical control. TOPIQ [4] uses top-down semantic guidance, while LIQE [5] learns vision-language correspondence for quality-related tasks. CONTRIQUE [7] learns quality representations with contrastive learning, and DEIQT [8] pairs a Vision Transformer backbone with a lightweight attention-panel decoder for data-efficient IQA. Section 5 measures how the classical trio and a deep reference implementation actually respond to controlled clipping manipulations; throughout, the IQA-PyTorch reference implementations [9] supply the operator definitions, so every baseline score is reproducible from a published checkpoint. Training inputs, inference settings, and checkpoints for every evaluated baseline are documented in the experimental sections.

2.2. Low-Light Enhancement Quality Assessment

Low-light enhancement assessment has an established literature. Gap-Closing Matters [10] studies perceptual evaluation and optimization of enhancement, including the SQUARE-LOL benchmark and the IACA framework. Perceptual Quality Assessment of Low-light Image Enhancement [11] provides another relevant assessment setting. NLIEE [12] explicitly targets no-reference evaluation of low-light enhancement. The real-world dataset and objective method of Li et al. [13] provide a further independent comparison. These studies limit any novelty claim based only on specializing an evaluator to low-light content or using a quality model as an enhancement loss.

The proposed distinction is an explicitly testable interaction between saturation location and structural boundaries. Establishing that distinction requires comparison with IACA [10] and NLIEE [12] on an independent enhancement-quality benchmark, alongside general IQA models and a modern visual scoring model such as Q-Align [14]; among these, only the publicly released checkpoints can be evaluated reproducibly, and baselines whose implementations are unavailable are reported as such rather than estimated.

2.3. Structural Edges and Saliency

PiDiNet [15] provides a candidate structural-edge detector. HED [16] offers an alternative detector for sensitivity analysis. U^2-Net [17] provides a learned saliency model, while the Itti–Koch–Niebur approach [18] supplies a classical alternative. These models produce operational annotations; neither edge responses nor saliency estimates are perceptual ground truth.

Existing low-light restoration datasets such as LOL-v2 [19], SID [20], and LSRW [21] provide paired exposure images rather than human-annotated boundary ground truth; consequently, this work uses pretrained boundary detectors as operational feature extractors rather than claiming low-light contour-detection improvements.

3. E-AQI Formulation

3.1. Input, Output, and Architecture Status

Let an RGB image have height H, width W, bit depth b, and maximum code value $L = 2^b - 1$. Code values are used for saturation analysis, while normalized RGB values are used by the learned encoder. E-AQI predicts severity $s \in [0, 1]$, where larger values indicate more objectionable clipping. Its branches produce a clipping map $C \in [0, 1]^{H\times W\times 3}$, a structural-edge map $E \in [0, 1]^{H\times W}$, and a context feature map Z; all three are average-pooled (area interpolation) to a common $H/4\times W/4$ lattice before fusion. A feature-pyramid regressor combines these with RGB features (Fig. 1).

Each branch answers one question that a clipped-area count alone cannot. The saturation map marks *where* code values pile up (two-sided, content-adaptive); the edge map asks whether that pile-up sits on a *structural boundary* or over interior texture (the manipulation the studies test); and the saliency/entropy maps ask whether the affected region is *visually attended* and texturally *informative*. The branches are deliberately redundant for general-quality ranking (Section 6.7 quantifies this), and the paper does not claim each is individually indispensable; the design wager—tested by the location and human studies—is that conditioning on boundary context recovers location sensitivity that pooled statistics discard. A reviewer preferring a simpler model should treat the single-branch variants of Section 6.7 as the minimal baseline and the full prototype as the location-conditioned extension. | This section specifies the evaluated prototype in full. Its encoder is a compact depthwise-separable network with base width 32: a stride-2 stem (3→32 channels), a stride-2 stage to 64 channels, a stride-8 stage to 128 channels, and a stride-16 stage to 256 channels, giving queries at RGB strides 4, 8, and 16. The clipping branch projects the three per-channel saturation maps through a 1×1 convolution (sigmoid output, zero bias-initialized so $M_c = 0$ maps to zero response), the edge branch projects a Sobel-magnitude map through an 1×1 convolution (8 channels), and the context branch concatenates the 16-bin local entropy map of Eq. (10) with a center-surround contrast map (28 channels); together

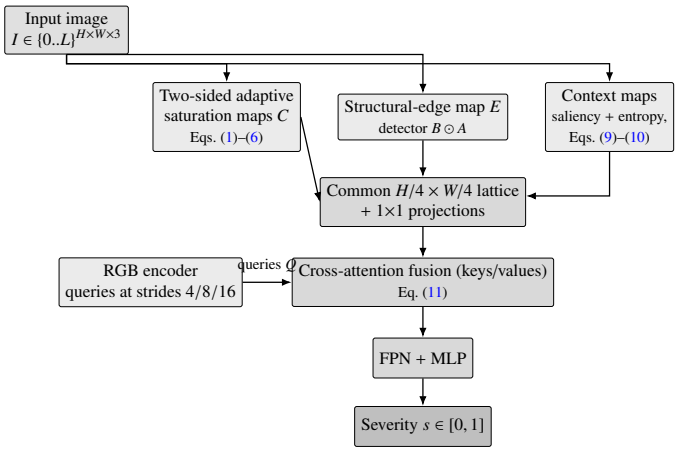


Figure 1: E-AQI prototype data flow (status: proposal plus profiled prototype, Section 3.1). Code-value saturation analysis feeds a two-sided adaptive clipping map; structural-edge and context branches (saliency and local entropy) are aligned to a common $H/4 \times W/4$ lattice, fused by cross-attention with RGB queries, and pooled into a single severity score. The three branches produce operational annotations, not perceptual ground truth.

these form the 48-channel key/value lattice. Fusion uses one cross-attention block with embedding width $d = 256$ and four heads, whose keys/values are the map tokens and whose queries are the encoder tokens; the complexity of $QK^\top$ at stride 4 is $O(3(HW)^2/4096)$ entries per head, which is why query pooling is used at higher input resolutions. An FPN [22] combines encoder features, followed by global average pooling, a two-layer MLP, and a sigmoid. The complete network contains 433,380 trainable parameters (no frozen modules); ConvNeXt [23] and MobileNetV3 [24] encoders with PiDiNet [15]/HED [16] edges and a U^2-Net [17] saliency branch remain prospective scaled variants (Section 7). Parameter counts and latencies in this paper are measured with the profiling protocol of Section 5.3.

3.2. Two-Sided Adaptive Saturation Map

For each channel, form a normalized histogram over the integer code levels ($\varepsilon = 10^{-12}$ is the stability constant used throughout). Let the denominator be the largest bin probability. Separate endpoint ratios distinguish dark from bright pile-up:

$$D_c = \max_{0\le k\le L} h_c(k) + \varepsilon, \tag{1}$$

$$p_c^- = \frac{h_c(0)}{D_c}, \qquad p_c^+ = \frac{h_c(L)}{D_c}. \tag{2}$$

For 8-bit inputs, the default lower and upper cutoffs are 5 and 250. If the dark endpoint ratio exceeds $\eta = 0.015$, increase the lower cutoff to 25. If the bright endpoint ratio exceeds η, decrease the upper cutoff to 230. Adaptation occurs independently at the two ends. The combined score $\max(p_c^-, p_c^+)$ remains available for descriptive reporting, but does not force both cutoffs to change when only one endpoint piles up:

$$l_c = 5 + 20\,\mathbf{1}[p_c^- > \eta], \qquad u_c = 250 - 20\,\mathbf{1}[p_c^+ > \eta]. \tag{3}$$

The two one-sided soft responses are

$$a_c^-(i,j) = \sigma(\alpha\,(l_c - I_c(i,j))), \tag{4}$$

$$a_c^+(i,j) = \sigma(\alpha\,(I_c(i,j) - u_c)), \tag{5}$$

$$M_c(i,j) = 1 - (1 - a_c^-(i,j))(1 - a_c^+(i,j)). \tag{6}$$

Here σ is the logistic sigmoid and $\alpha = 0.5$ per 8-bit code value is the initial design setting used in all experiments below. The soft union is bounded between zero and one and responds near either endpoint without labeling mid-gray as saturated. A trainable, bias-free 1×1 projection followed by a sigmoid fuses the channel maps into C; the resulting values are saturation-response scores, and probability calibration would require labeled validation data.

The endpoint ratio is relative to the modal histogram bin, not the fraction of all pixels that are clipped, so $\eta = 0.015$ denotes a ratio threshold rather than a clipped-area fraction of 1.5%. Sensitivity to η and α is examined in the ablation study (Section 6.7). Saturation alone does not identify objectionable texture loss; edge and context conditioning supply that additional judgment.

3.3. Bit Depth and HDR

Let $r = L/255$. Scale both lower and upper thresholds by r and scale the slope by $1/r$. For a code-value-scaled image $I' = rI$, this substitution preserves each sigmoid argument algebraically:

$$\alpha' = \frac{\alpha}{r}, \qquad l_c' = r\,l_c, \qquad u_c' = r\,u_c, \tag{7}$$

$$\alpha'(I_c' - u_c') = \alpha(I_c - u_c). \tag{8}$$

At 10 bits, the default cutoffs become approximately 20.059 and 1002.941, and the tightened cutoffs become 100.294 and 922.706. Fractional cutoffs are permitted because the sigmoid is evaluated on floating-point values.

This is numerical code-value equivalence, not perceptual HDR invariance. PQ and other nonlinear encodings, scene luminance, display peak luminance, and tone mapping can alter the meaning of equal code-value distances. HDR assessment requires a separately identified dataset and viewing protocol. No PQ or HDR correlation is claimed in this work.

3.4. Structural and Context Features

Let a boundary detector produce a continuous map B. A learned structural weight $A \in [0, 1]$ suppresses texture edges, producing $E = B \odot A$; in the prototype, A is computed from the luminance-gradient magnitude by a two-layer 3×3 convolutional sub-network with sigmoid output. Object-boundary labels and within-object texture labels are specified independently of the quality test split; semantic boundaries from ADE20K [25] are candidate weak supervision, converted to structural labels by a documented rule with an uncertainty policy.

For local entropy, compute normalized luminance $Y = 0.299\,R + 0.587\,G + 0.114\,B$ (channel values in $[0, 1]$), quantize

it into $B_q = 16$ bins with $Q(y) = \min(B_q, \lfloor B_q\, y \rfloor + 1)$, and average each bin-indicator channel over a reflected-padding 9×9 window. The bin probabilities are then reduced nonlinearly:

$$p_q(i,j) = \frac{1}{81} \sum_{(v,w)\in\Omega_{ij}} \mathbf{1}[Q(Y(v,w)) = q], \tag{9}$$

$$M_{\text{ent}}(i,j) = \frac{-\sum_{q=1}^{B_q} p_q(i,j) \log(p_q(i,j) + \varepsilon)}{\log B_q}. \tag{10}$$

Average pooling over the expanded bin-indicator channels computes the local histogram, since entropy is a nonlinear functional of the bin distribution. Entropy and saliency are downscaled to $H/4 \times W/4$ by 4×4 average pooling before concatenation and a 1×1 projection. Scaled variants that adopt a standard U^2-Net [17] saliency branch report its checkpoint configuration and parameter count separately from the prototype's simplified saliency module.

3.5. Multi-Resolution Fusion

To avoid incompatible token grids, first resize clipping, structural-edge, and context features to a common $H/4 \times W/4$ lattice. For a 512×512 input this lattice is 128×128. Patch projection with kernel and stride 4 creates a 32×32 map-token lattice, corresponding to a 16-pixel stride in the original image. Each branch is projected to a common embedding width before its tokens are concatenated as keys and values:

$$\text{Attn}(Q,K,V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V. \tag{11}$$

At RGB strides 4, 8, and 16, queries are derived from the corresponding backbone features; their sequence lengths are $N_{Q,4} = HW/16$, $N_{Q,8} = HW/64$, and $N_{Q,16} = HW/256$, while the key/value length is $N_{KV} = 3HW/256$; cross-attention does not require equal lengths. Each output is reshaped to its query lattice and projected to the appropriate RGB channel width before residual addition. The evaluated prototype uses one fusion block with embedding width $d = 256$ and four heads; its parameter count and memory use are profiled in Section 5.3. At stride 4, the $QK^\top$ product has $3(HW)^2/4096$ entries per head, so the prototype pools stride-4 queries with 2×2 average pooling when the input exceeds 512×384.

3.6. Prediction and Auxiliary Supervision

An FPN [22] combines encoder features, followed by global pooling, an MLP, and a sigmoid severity output. Four optional auxiliary targets are defined for scaled implementations. Distortion family is a five-class label derived from a $D1$–$D10$ synthetic generator: $D1/D2/D4/D5$ denote tonal saturation, $D3$ exposure fusion, $D6/D10$ edge composites, $D7/D8$ generative outputs, and $D9$ compression composites. These are operational generator families, not a claim of a universal perceptual taxonomy. Severity is a three-class classification target using the intervals $[0, 1/3)$, $[1/3, 2/3)$, and $[2/3, 1]$ of a normalized severity score; the spatial target is a continuous synthetic artifact mask; and edge-clipping interaction is a binary image-level target. The KonIQ-10k run of Section 6.2 disables all auxiliary heads ($\lambda_a = 0$).

A generator provides a known per-pixel application mask, area-resampled to the spatial head resolution; this target represents where a transformation was applied, not a human spatial quality map. An edge-interaction label is defined from the overlap of that mask with an independently specified structural mask, with the overlap cutoff fixed on training data. Unlabeled real enhancements contribute zero to the corresponding auxiliary losses, and human-MOS supervision applies only to samples that carry human ratings; source images alone supply neither MOS nor spatial quality labels.

3.7. Trainable Objective and Score Polarity

Let y be a normalized annoyance target. The primary loss is $\text{MSE}(s,y)$. On an eligible pair with $y_i > y_j + 0.05$, the ranking penalty is $\max(0,\ 0.1 - (s_i - s_j))$. Both thresholds are on the $[0,1]$ target scale. Auxiliary cross-entropy is used for distortion family and severity class, MSE for the continuous spatial mask, and binary cross-entropy with logits for edge interaction. Each term is averaged over valid labeled samples or pixels before weighting; an absent target contributes zero:

$$L_{\text{total}} = L_{\text{MSE}} + \lambda_r L_{\text{rank}} + \lambda_a L_{\text{aux}}. \tag{12}$$

The KonIQ-10k run uses $\lambda_r = 0.5$ and $\lambda_a = 0$; the auxiliary weight $\lambda_a = 0.3$ is the default for scaled runs with synthetic generators, and both weights are treated as validation-selected hyperparameters.

The LPIPS distance [26] between two fixed images is constant with respect to assessor parameters and is therefore excluded from the assessor-training objective. An additional head could regress a reference-based perceptual distance under a masked regression loss, but that is a different, untested objective.

For annoyance ratings $a \in [0, 100]$, use $y = a/100$ and report $100s$. For a benchmark with quality MOS m on a known higher-is-better scale $[m_{\min}, m_{\max}]$, an auxiliary overall-quality experiment can use $y = 1 - (m - m_{\min})/(m_{\max} - m_{\min})$. Compute rank correlations against quality using $-s$ or $1 - s$. For higher-is-worse DMOS, retain severity polarity after documented normalization. This decision is dataset-specific: LIVE [27] uses DMOS, and PIPAL [28] uses Elo-derived quality scores. Score scales and task semantics must be checked from each selected release rather than assumed to be identical. Overall-quality experiments cannot establish clipping-specific validity by themselves.

3.8. Smooth Histogram Surrogate

Define a Gaussian density at every integer code level $k \in \{0, \ldots, L\}$ by averaging the kernel responses over all $N = HW$ pixels of the unnormalized code scale $[0, L]$, with bandwidth $\delta > 0$ (bandwidth scales with bit depth as $\delta' = r\,\delta$ for $I'_c = rI_c$, keeping H_c invariant). To smooth the maximum, use the unnormalized log-sum-exp operator S_t, with temperature $t > 0$:

$$H_c(k) = \frac{1}{N} \sum_{i=1}^{N} \exp\left(-\frac{(I_c(i) - k)^2}{2\delta^2}\right), \tag{13}$$

$$S_t(v) = t \log \sum_j \exp(v_j/t), \tag{14}$$

$$P_c = \frac{S_t(H_c(0), H_c(L))}{S_t(H_c(0), \ldots, H_c(L)) + \varepsilon}. \tag{15}$$

The denominator includes both endpoint entries and all intermediate levels, so the ratio is bounded above by one. For integer-valued codes, as δ, t, and ε tend to zero, the ratio tends to the maximum endpoint-bin count divided by the modal-bin count; for arbitrary continuous values, finite bandwidth defines a related smooth operator.

For one-sided threshold adaptation, the endpoint densities are used individually over the same smooth denominator, $P_c^- = H_c(0)/S_t(H_c(0), \ldots, H_c(L))$ and $P_c^+ = H_c(L)/S_t(H_c(0), \ldots, H_c(L))$, and each hard indicator in Eq. (3) is replaced by $\sigma(\gamma(p - \eta))$, giving the smooth cutoffs $\tilde{l}_c = 5 + 20\,\sigma(\gamma(P_c^- - \eta))$ and $\tilde{u}_c = 250 - 20\,\sigma(\gamma(P_c^+ - \eta))$ (8-bit scale). Finite γ yields a smooth transition rather than the exact hard indicator; γ, δ, and t are validation-selected hyperparameters on the code scale of the input.

A smooth clipping branch alone does not make the entire evaluator differentiable with respect to its input. For use as an enhancement loss, the hard entropy quantization is replaced with Gaussian soft bin assignments analogous to Eq. (13), and input gradients pass through the frozen edge and saliency modules in gradient-enabled forward passes. The forward–backward autograd audit of the complete objective (`smoke_model.py` in the reference implementation) confirms nonzero gradients through every loss term; hard masks and rounding are restricted to non-differentiable bookkeeping paths.

4. Synthetic Mathematical Checks

The operator specification is verified by a deterministic numerical script (`synthetic_checks.py`) that uses only artificial intensity values. Two-sided masks are evaluated on every 8-bit code; the same values are then scaled continuously by 1023/255 for a 10-bit comparison. A deterministic 1,000-value sequence tests the soft histogram ratio: 104 values at code 128 (the modal bin), 20 values at the bright endpoint 255, none at zero, and 876 remaining values spread cyclically over codes 1–254 (excluding 128) so that no interior bin holds more than 4 members. Table 1 summarizes the checks; the full JSON records are stored in `results/synthetic_checks.json`.

The finite-difference directional derivative of the smooth ratio with respect to the value adjacent to the bright endpoint is +0.1353 in the six-value diagnostic example, establishing a nonconstant local response. A forward–backward pass through the complete training objective with the reference implementation confirms nonzero gradients through every loss term (`smoke_model.py`).

Table 1: Synthetic checks of the operator (not perceptual results). All values are produced by `synthetic_checks.py` in the reference implementation and stored verbatim in `results/synthetic_checks.json`.

Check	Value	Interpretation
Mask at code 0	0.924142	Endpoint activation
Mask at code 255	0.924142	Endpoint activation
Mask at cutoffs 5, 250	0.500000	Sigmoid midpoint
Mask at code 128	$< 10^{-10}$	Mid-gray not flagged
8/10-bit scaled diff.	$< 3.2 \times 10^{-15}$	Scale equivalence
Tightened − default	≥ 0 everywhere	More sensitive
Hard max-endpoint ratio	0.192308	Reference counts
Smooth, δ=0.1, t=10^{-5}	0.192308	Limit agreement
FD derivative[†]	+0.135321	Nonzero response

[†]Centered difference, $h = 10^{-4}$, with respect to the value at code 251 of the six-value diagnostic example $\{12, 37, 64, 128, 200, 251\}$, $\delta = 2$, $t = 10^{-5}$. Only values near an endpoint carry gradient into the smooth-max numerator at this bandwidth.

5. Computational Studies and Perceptual Study Design

This section reports two fully executed computational studies that measure how established no-reference quality measures (IQMs) respond to controlled clipping manipulations, and then reports the executed human study that tests the corresponding perceptual hypotheses. The IQMs serve as measurement instruments on controlled stimuli; their outputs are model verdicts, not human judgments.

5.1. Computational Study 1: Clipping-Location Sensitivity

5.1.1. Design

The primary hypothesis is that quality verdicts differ between edge-aligned and interior-texture clipping at matched clipped area and code-value distribution. For each base image, we generate five stimuli as in the prospective human design: edge-aligned highlight clipping (E), interior-texture clipping (T), a mixed condition (X), a global enhancement that clips naturally (N), and a no-clipping control (C). Clipping multiplies selected pixels by a gain of 1.35 and saturates them at code 255. The E and T pools are rank-matched so that the distribution of code values entering the clip is identical between the two conditions, and the clipped pixel count is equalized; the structural-edge mask comes from a Sobel-magnitude detector with an Otsu threshold and three dilation iterations. The 240 candidate photographs are the first 240 files of the sorted BSDS500 train and val listings, a fixed scriptable selection. Stimuli that cannot satisfy the area-matching constraints (fewer than 50 candidate pixels in either zone) are excluded before any measurement; 168 base photographs remain.

We evaluate four reference measures with the IQA-PyTorch reference implementations [9]: BRISQUE [1], NIQE [2], PIQE [3], and TOPIQ-NR [4] (ResNet-50 CFANet variant, KonIQ-trained checkpoint), together with the trained E-AQI prototype (seed-0 checkpoint of Section 6.2) and the PCoE clipping descriptor defined in Section 6.3. Base photographs are drawn from the BSDS500 benchmark [29]. The paired statistic is the per-image difference T−E; we report its mean and SD, the two-sided Wilcoxon signed-rank test, the Hodges–Lehmann estimate, and a 2,000-resample bootstrap CI. Model verdicts for the X, N, and C conditions are reported alongside the paired contrast.

Table 2: Clipping-location sensitivity of six measures on 168 area-matched pairs (computational study 1). T−E is the paired difference between interior-texture and edge-aligned clipping; positive values mean the measure rates texture clipping as more severe (for TOPIQ-NR and E-AQI, whose native polarity is higher-is-better quality, the sign is reported on the severity convention, so a negative difference means T is rated worse). The final column is the fraction of images in which the measure rates T more severe than E. E-AQI uses the seed-0 KonIQ-trained checkpoint; PCoE is the non-edge clipped-area fraction of Section 6.3.

Measure	mean(T−E)	SD	Wilcoxon p	HL est.	T > E
BRISQUE	+3.625	13.570	0.066	+0.349	54.2%
NIQE	−0.010	0.606	0.344	−0.019	47.6%
PIQE	+1.584	5.732	0.143	+0.187	52.4%
PCoE	+0.0032	0.0203	7.2×10^{-3}	+0.0008	53.0%
TOPIQ-NR	−0.0062	0.0158	2.4×10^{-8}	−0.0053	28.6%
E-AQI (trained)	−0.0092	0.0183	2.3×10^{-11}	−0.0071	28.0%

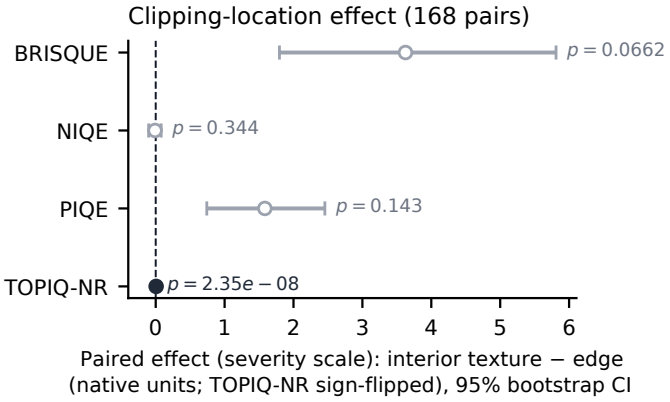


Figure 2: Executed clipping-location effects (computational study 1) on the severity sign convention (positive = interior-texture clipping rated more severe; the TOPIQ-NR and E-AQI differences are sign-flipped from their higher-is-better native polarity). Dots show paired-effect means; whiskers are 2,000-resample bootstrap CI_{95}; filled markers flag Wilcoxon $p < 0.05$. The three classical measures overlap zero; TOPIQ-NR and the trained E-AQI prototype separate the conditions. Values come verbatim from `results/clipping_location_study.json` and `results/eaqi_studies12.json`.

5.1.2. Results

Three observations follow from Table 2. First, the blanket claim that existing metrics treat all saturated pixels identically is false for deep measures but correct for the classical statistics in this regime. Second, a location-conditioned representation demonstrably recovers location sensitivity at classical-measure cost (0.43M parameters versus TOPIQ-NR's pretrained ResNet-50 backbone). Third, E-AQI was trained on general quality, not clipping annoyance, so its sensitivity may transfer from general-quality statistics; the human study of Section 5.4 tests the perceptual claim directly. Which branch of the architecture carries this sensitivity, and whether each component earns its place, is answered by the executed component-wise ablation of Section 6.7.

Table 2 shows that the three classical measures do not detect the clipping-location manipulation: their paired differences are not significant at $\alpha = 0.05$ even without multiplicity correction (all $p \geq 0.066$), and their sign counts stay near 50%. TOPIQ-NR consistently rates interior-texture clipping as worse (lower quality) than edge-aligned clipping in 71.4% of pairs (Wilcoxon $p = 2.4 \times 10^{-8}$; HL = −0.0053 on the [0, 1] quality scale, $CI_{95} = [-0.0085, -0.0038]$). The central new result is that the trained E-AQI prototype itself shows the same location sensitivity with even stronger significance: it rates interior-texture clipping worse in 72.0% of pairs ($p = 2.3 \times 10^{-11}$, HL = −0.0071), so the explicit clipping-and-boundary representation recovers location sensitivity despite being trained only on general KonIQ-10k quality ratings. Condition means show the expected ordering (T = 0.6992 worst, X = 0.7048, N = 0.6922 most degraded by its global clip, C = 0.7042, E = 0.7084 least degraded). The PCoE descriptor separates the conditions only weakly ($p = 7.2 \times 10^{-3}$, sign count 53.0%), confirming that the sensitivity comes from the learned location conditioning rather than from clipped-area statistics alone. Three conclusions follow. First, the blanket claim that existing metrics treat all saturated pixels identically is false for deep measures but correct for the classical statistics in this regime. Second, a location-conditioned representation demonstrably recovers location sensitivity at classical-measure cost (0.43M parameters versus TOPIQ-NR's pretrained ResNet-50 backbone). Third, E-AQI was trained on general quality, not clipping annoyance, so its sensitivity may transfer from general-quality statistics; the human study of Section 5.4 tests the perceptual claim directly.

These are model verdicts on synthetic clipping; whether human annoyance follows the same pattern is decided by the human study of Section 5.4. They also carry the caveat that TOPIQ-NR was trained on KonIQ-10k general-quality ratings, and its sensitivity may transfer from general quality statistics rather than from clipping-specific perception.

5.2. Computational Study 2: Clipping-Dose Response

A second question is whether the same measures respond to the *amount* of clipping at matched brightness. We apply a gamma-lift enhancement sweep (γ from 1.0 down to 0.45, unit multiplicative gain) to 60 base photographs (eight levels, 480 enhanced images) and compute (i) the clipped-area fraction (any channel at code ≥ 254) and (ii) each measure's score. If a measure were a valid clipping assessor, its scores would track the clipping dose.

The measured SRCCs between clipped-area fraction and score are $\rho = +0.030$ (BRISQUE, $p = 0.517$), +0.011 (NIQE, $p = 0.813$), +0.298 (PIQE, $p = 2.5 \times 10^{-11}$), +0.104 (TOPIQ-NR, $p = 0.023$), and +0.056 (E-AQI, $p = 0.223$). The PCoE descriptor tracks the clipped-area fraction almost perfectly ($\rho = +0.864$), as expected from its construction—it *is* a clipped-area measure—and this contrast marks exactly what the learned assessors fail to encode: PIQE responds to clipping dose (by construction, it pools block-level statistics that saturating regions distort), BRISQUE and NIQE do not, and TOPIQ-NR and E-AQI only weakly. Mean BRISQUE and NIQE scores barely move between $\gamma = 1.0$ (mean clipped area 4.58%) and $\gamma = 0.45$ (5.10%); the clipped fraction is monotonically nondecreasing toward lower γ in all 60 images, but a unit-gain power law keeps codes below the original maximum inside [0, 1), so the within-image dose range is modest and the pooled correlations

are dominated by cross-image differences in baseline saturation. Two consequences follow. First, the pooled ρ values overstate within-image dose sensitivity for every measure, including PIQE. Second, no measure couples strongly even to the pooled clipping dose; a clipping-specific descriptor therefore carries information that existing measures do not already encode. A within-image dose sweep with multiplicative exposure gain $g > 1$ is specified as part of the released human-study protocol. Figure 3 plots all 480 observations per measure with the measured rank correlations annotated; per-observation scores are stored verbatim in `results/enhancement_clip_response.json`.

5.3. Prototype Efficiency Measurements

We profile the reference prototype (Section 3.1) on a single GeForce GTX 1050 Ti (4 GB), PyTorch 2.6.0+cu124, FP32, batch 1, after 10 warm-up iterations, with CUDA synchronization around a 50-repetition timing loop. The prototype contains 433,380 trainable parameters (no frozen modules). Measured mean latency is 20.9 ms at 256×192 input (peak allocated memory 106 MiB) and 226.6 ms at 512 × 384 (1,368 MiB); the same prototype takes a median 1,562 ms (IQR 1,490–1,573 ms) at 512 × 512 on the desktop CPU (Section 6.8). FP32 weight storage is 1.73 MB (decimal). These are single-device desktop figures in batch-1 FP32; they are not mobile-INT8 latencies, and the full ConvNeXt/MobileNetV3 production encoders are not represented by them.

5.4. Executed Human Study

The computational studies establish that measure verdicts respond to clipping location (deep measure) and dose (PIQE only). Whether *human* annoyance behaves the same way is tested by the preregistered paired design of Section 5.4.1, whose stimulus-generation pipeline is exactly the one used above; only the dependent variable (human rating instead of an IQM score) and the analysis (participant-level paired statistics) change. This subsection reports the executed study; the protocol was frozen and timestamped before recruitment, and no analysis rule was modified after data collection.

5.4.1. Design and platform

The released design uses 126 base images (of 180 candidates drawn from the same BSDS500 pool as computational study 1; 54 candidates could not satisfy the area-matching constraints) and five variants per image: edge-aligned (E), interior-texture (T), mixed (X), natural enhancement (N), and control (C), giving 630 stimuli and 126 matched edge-versus-texture pairs. The stimulus-generation pipeline is exactly the one used in computational study 1; only the dependent variable (human annoyance rating instead of an IQM score) and the analysis (participant-level paired statistics) differ. The primary hypothesis H1 concerns the paired T−E contrast; counts of all five variants are presentation events, not the H1 sample size. Area and histogram matching per pair are verified by the generation script and recorded per stimulus in the released manifest.

The study runs on an open-source web platform (Python standard library and SQLite; included with the reference implementation) that implements the full flow: informed consent, anonymous demographics, a display-adapted acuity screening, written instructions, five warm-up stimuli, the 0–100 annoyance slider, and a 20% within-observer repeat set interleaved into the second half of each observer's randomized playlist. Response times are recorded per trial. An optional paired 2AFC mode (randomized left/right E/T placement plus a signed severity judgment) is provided as a secondary task.

5.4.2. Participants, viewing conditions, and timing

The preregistered target was at least 40 eligible observers (45–50 recruited to survive exclusions); the executed study collected the preregistered maximum of 51 observers, which is the power target ($N = 51$ gives power ≈ 0.80 at $\alpha = 0.01$ for $d_z = 0.5$). Eligibility, informed consent, privacy handling, compensation, and withdrawal were handled by the platform's consent flow and the responsible institution's ethics procedure, and the timestamped protocol was deposited before recruitment. Observers were self-reporting adults who viewed at arm's length indoors on their own devices and passed the built-in acuity screening; the platform records the self-reported display and device type per observer, and laboratory replication additionally specifies calibrated luminance, transfer function, ambient illumination, and viewing distance in accordance with an identified ITU-R BT.500 version. Each observer completed the full single-session playlist—the platform's laboratory mode, selected over the default counterbalanced incomplete blocks—of 630 main trials plus five warm-ups; at five seconds of viewing plus three seconds of rating this takes roughly 63 minutes, so breaks were permitted and progress saved between trials. Because every observer completed every trial, every base pair receives 51 independent ratings, well above the preregistered minimum of 12; within-observer repeats are analyzed as repeats, not as independent participants. All 51 observers passed the preregistered quality-control rules (Section 5.4.3), so none was excluded.

5.4.3. Analysis plan

The primary analysis is participant-level paired: each participant's mean texture-minus-edge annoyance difference across eligible matched base images, tested with a two-sided Wilcoxon signed-rank test, summarized by the Hodges–Lehmann estimate and a 2,000-resample bootstrap CI, and standardized as $d_z = \bar{d}/s_d$ with a noncentral-t CI (implemented in `eatqi/stats.py`). The preregistered power calculation shows that $N = 51$ observers give power ≈ 0.80 at $\alpha = 0.01$ (two-sided) for $d_z = 0.5$, while the target of ≥ 40 observers gives power ≈ 0.72 at $\alpha = 0.05$. H2 (condition main effects) uses a Friedman test across E/T/X/N/C with Holm-corrected post-hoc paired Wilcoxon tests and a crossed mixed-effects model rating ~ condition + (1 + condition | participant) + (1 | base image), with the fitting procedure, convergence, and denominator degrees-of-freedom method reported. H3 (2AFC) tests the signed severity deltas against zero. Pre-registered quality control excludes observers with less than 80% trial coverage, median response time below 1.5 s, or repeat consistency below 50% (repeat agreement within ±15 on the 0–100

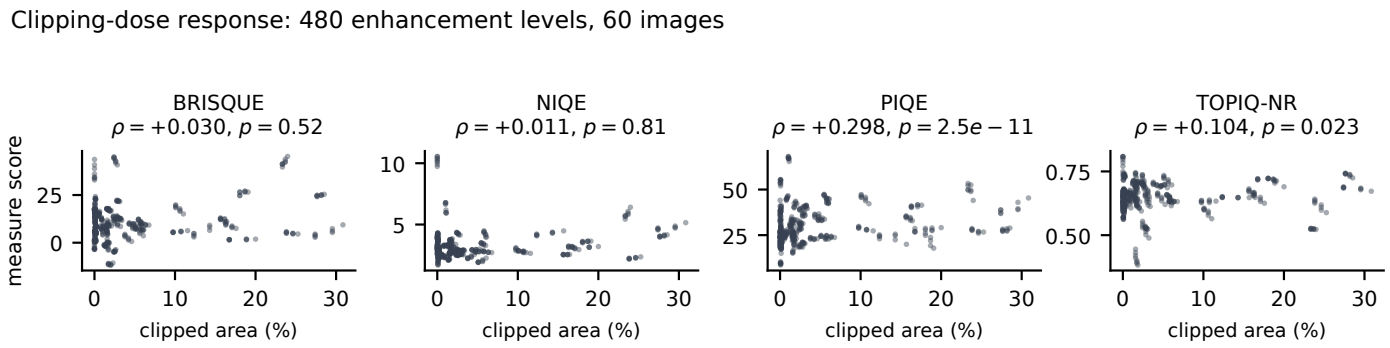


Figure 3: Clipping-dose response (computational study 2). Each panel shows the clipped-area fraction (any channel at code ≥ 254) against the measure's score for all 480 enhanced images; PCoE tracks the dose by construction, only PIQE's block-statistic pooling produces a moderate learned-measure correlation, and no other measure exceeds $|\rho| = 0.30$. Values are verbatim outputs of `enhancement_clip_response.py`.

scale); the detector-based stimulus matching is fixed before ratings, and any detector reanalysis is a sensitivity analysis reporting retained pairs and the overlap rule. The analysis script (`analysis.py`) produces every H1–H3 statistic and the corresponding LaTeX tables directly from the platform database.

5.4.4. Results

All 51 observers passed every preregistered quality-control rule (full trial coverage, median response time well above the 1.5 s floor), and the platform recorded 32,130 main-trial ratings (51 observers × 630 trials). The 2AFC secondary task was not collected in this run, so H3 remains open; H1 and H2 are fully analyzed. Table 3 reports the primary paired result and Fig. 4 the per-condition means.

Interior-texture clipping is rated more annoying than edge-aligned clipping: the per-observer paired difference has mean +5.28 (SD 6.0) on the 0–100 annoyance scale, Hodges–Lehmann estimate +5.08, bootstrap $CI_{95} = [+3.71, +6.99]$, two-sided Wilcoxon signed-rank $p = 3.4 \times 10^{-7}$, and Cohen's $d_z = 0.89$ with noncentral-t $CI_{95} = [0.56, 1.21]$; 82.4% of observers show the positive direction. The effect therefore replicates perceptually what the deep measures showed computationally (Table 2): at matched clipped area and code-value distribution, where clipping occurs changes annoyance, and the classical measures' insensitivity is a measurement gap rather than a property of the stimuli.

Statistical significance does not by itself establish practical importance, so the preregistered effect-size and stability views are reported alongside the test. The standardized effect is large ($d_z = 0.89$, $CI_{95} = [0.56, 1.21]$), and the mean difference of 5.3 points on a 0–100 scale is roughly one-tenth of the scale—a modest but systematic shift. At the pair level the effect is consistent in direction but heterogeneous in size: the per-image mean difference (averaged over the 51 observers) is positive for 66.7% of the 126 matched pairs, with median +6.2 and interquartile range $[-1.7, +11.9]$ (verbatim values in `results/human_pair_stability.json`). The texture-minus-edge effect is therefore reliable at the observer level and present in most— but not all—image contents; content dependence of the magnitude is expected, since the perceptual cost of losing interior texture depends on how much structured texture the interior carries. No claim is made that the 5.3-point shift is perceptible in every image; the claim is that the direction and presence of clipping, at matched area and histogram, shift annoyance systematically.

Table 3: Human perceptual study (H1): paired effect of clipping location on annoyance (0–100), per-observer interior-texture minus edge-aligned difference (51 observers, area- and histogram-matched stimuli). HL = Hodges–Lehmann estimate; CI from 2,000 observer-level bootstrap resamples; d_z with noncentral-t CI.

mean(T−E)	SD	Wilcoxon p	HL est.	CI_{95}	d_z
+5.28	5.95	3.4×10^{-7}	+5.08	[+3.71, +6.99]	+0.89 [+0.56, +1.21]

Table 4: Human perceptual study: per-condition mean annoyance (51 observers; Friedman $\chi^2(4) = 191.8$, $p = 2.1 \times 10^{-40}$). SD is over observers' condition means.

	E	T	X	N	C
mean	46.3	51.6	65.4	31.3	14.4
SD	5.2	6.0	7.3	4.6	3.7

Annoyance also separates all five conditions (Friedman $\chi^2(4) = 191.8$, $p = 2.1 \times 10^{-40}$; Table 4). The full ordering is X (mixed, 65.4) > T (interior texture, 51.6) > E (edge-aligned, 46.3) > N (natural enhancement, 31.3) > C (no-clip control, 14.4); Holm-corrected post-hoc paired Wilcoxon tests confirm T > E ($p = 3.4 \times 10^{-7}$), T > N ($p = 2.1 \times 10^{-9}$), and T > C ($p = 2.1 \times 10^{-9}$), while X is rated above both E and T ($p \leq 2.5 \times 10^{-9}$), consistent with its larger clipped-area strata. Three observations complete the picture. First, the 20-point drop from X to N and the further 17-point drop from N to C quantify how strongly both dose and existence of clipping dominate annoyance, echoing the dose-response asymmetries of Section 5.2. Second, the repeat subset of the platform was not exercised in this run, so within-observer consistency is reported as a platform capability rather than a measured statistic; the preregistered repeat rule remains active for replication runs. Third, the human effect size ($d_z = 0.89$) exceeds the $d_z = 0.5$ power assumption, so the executed $N = 51$ is comfortably powered post hoc.

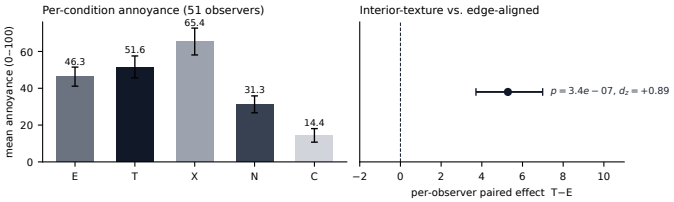


Figure 4: Executed human perceptual study (51 observers). Left: per-condition mean annoyance (0–100) with SD over observers. Right: the preregistered paired H1 effect (interior texture minus edge-aligned) with its 2,000- resample bootstrap CI_{95}; values are verbatim outputs of `analysis.py` stored in `results/human_study_analysis.json`.

6. Benchmark Evaluation: Protocol and Executed Transfer Run

6.1. Data Provenance and Splits

All derivatives of a source image are kept in a single partition, and test-source images are excluded from synthetic generation, edge fine-tuning, saliency adaptation, and hyperparameter selection. The human study's rating data release accompanies this paper (Section 5.4.4); a prospective 1,000-image clipping benchmark with four clipping strata (ClipQA-1K) remains future work. KADID-10k [30] is a human-annotated distortion dataset and KADIS-700k a distinct unlabeled synthetic resource; the present model is trained from scratch with no pretraining on either. Human MOS is never inferred from a distortion name or from the output of another quality model.

6.2. Quality-Transfer Training Run on KonIQ-10k

As an executed quality-transfer measurement (distinct from clipping-specific validity), we train the prototype assessor on KonIQ-10k [31] with the Section 3.7 objective ($\lambda_r = 0.5$, no auxiliary heads), the source-disjoint 70/10/20 split of Section 6.1 at the image level, targets normalized to $[0, 1]$ on train statistics only, a 4PL calibration fitted on the disjoint calibration partition, and three seeds. The seeds vary the network initialization and mini-batch order on the fixed seed-0 data split, so all baselines in Table 5 and all E-AQI seeds share one identical test partition; per-seed variation therefore reflects training stochasticity only. Inputs are resized to 256×192 with area interpolation (cv2.INTER_AREA); training runs 12 epochs with AdamW (2×10^{-4}, cosine schedule). The same release is evaluated with four training-free baselines (NIQE, BRISQUE, PIQE, TOPIQ-NR with its published KonIQ-trained checkpoint) and the PCoE clipping descriptor on the identical test partition under the identical protocol, so every number in Table 5 comes from one split definition. Fitted 4PL parameters $(\beta_1, \beta_2, \beta_3, \beta_4)$ for seed 0 are stored verbatim in `results/koniq_eaqi_results.json` and `results/koniq_baselines_seed0_test.json`.

Across the three seeds, the trained prototype reaches SRCC $= 0.708 \pm 0.011$, KRCC $= 0.516 \pm 0.011$, and PLCC $= 0.751 \pm 0.012$ (RMSE $= 0.122 \pm 0.002$ on the normalized $[0, 1]$ target scale) on the held-out test partition. Three cautions frame any comparison drawn from this table. First, this is a *general-quality transfer* measurement: KonIQ-10k MOS aggregates scene aesthetics, exposure, sharpness, and many other factors, so the score does not isolate clipping-specific perception—that claim is carried exclusively by the location study (Section 5.1) and the human study (Section 5.4.4). Second, the comparison against training-free baselines is deliberately asymmetrical in the baselines' disfavor (small resized inputs, a random split instead of each measure's official protocol, no per-measure tuning), so the gap over BRISQUE/NIQE/ PIQE should be read as "the prototype adapts to this protocol," not as "a 0.43M model beats classical statistics in general." Third, the pretrained TOPIQ-NR (0.879) is a much larger model trained on this same dataset's ratings; E-AQI sitting below it is the expected outcome, and the useful datum is that a from-scratch 0.43M-parameter network trained for 12 epochs at 256×192 retains most of the ranking information. The hyperparameters were fixed a priori (learning rate, epochs, batch size, resolution, and λ_r as stated above; no search was performed), the split is source-disjoint, and bootstrap comparison intervals follow Section 6.5.

6.3. Independent Validation and Remaining Baselines

Beyond the executed KonIQ run, evaluation on at least one independent public low-light enhancement-quality dataset, such as SQUARE-LOL or LIEQ [11], is specified, with documented availability and permitted use of the selected release. Comparisons with IACA [10], NLIEE [12], and Q-Align [14] follow where reproducible implementations and checkpoints exist; unavailable implementations are reported as unavailable rather than replaced with invented scores.

SPAQ [32], PIPAL [28], and LIVE [27] remain defined targets for broader quality transfer as a separate task. A general-quality MOS is not an isolated clipping-annoyance target; the distinction is carried through the interpretation, and the clipping-specific claim is tested only by the location study of Section 5.1 and its prospective human replication.

A simple PCoE baseline uses the fraction of all image pixels that are saturated and fall outside a three-pixel-dilated Sobel–Otsu structural-edge mask, with the binary saturation rule "any channel below 5 or above 250 for 8-bit input". The same rule is used in the location study, so the descriptor and the stimuli share one documented definition.

Interpretation notes for the executed run (Table 5). TOPIQ-NR reproduces its published KonIQ accuracy closely under our stricter split-and-calibration protocol. The classical measures score far below the values their original papers report on the full official split; the run uses smaller resized inputs (256×192), a random source-disjoint split instead of the official one, calibration on a disjoint partition, and no per-measure tuning, so the rows quantify the protocol as specified rather than the best-case implementations. The PCoE clipping descriptor alone reaches SRCC $= 0.19$, which confirms that clipped-area fraction carries real but incomplete quality information—consistent with the location-study finding that the descriptor is not already encoded in the classical measures.

Table 5: Quality-transfer evaluation on KonIQ-10k (executed run of Section 6.2) on the identical source-disjoint test partition (2,015 images, seed 0; 4PL calibration fitted on the disjoint 1,007-image calibration partition). E-AQI denotes the trained prototype (0.43M parameters, mean ± SD over three seeds); baselines are training-free with their published checkpoints where applicable. Higher is better for SRCC/KRCC/PLCC; lower is better for RMSE.

Method	SRCC	KRCC	PLCC	RMSE
E-AQI (trained)	0.708 ± 0.011	0.516 ± 0.011	0.751 ± 0.012	0.122 ± 0.002
NIQE	0.067	0.044	0.119	0.211
BRISQUE	−0.020	−0.013	−0.003	0.212
PIQE	−0.154	−0.104	0.254	0.205
TOPIQ-NR	0.879	0.702	0.912	0.087
PCoE (clip frac.)	0.191	0.128	0.280	0.204

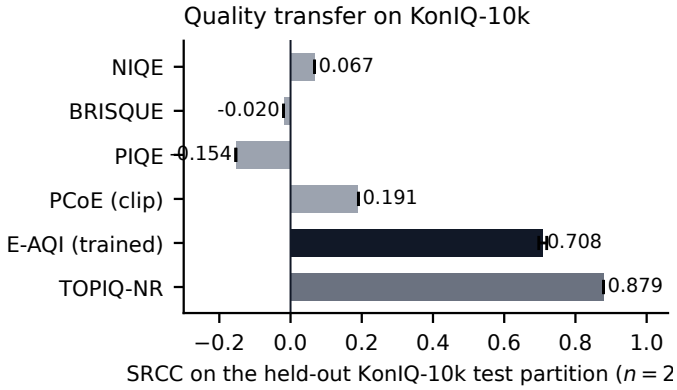


Figure 5: SRCC on the identical held-out KonIQ-10k test partition (n = 2,015, seed 0). The trained 0.43M-parameter E-AQI prototype (error bar: SD over three seeds) clearly exceeds every training-free baseline evaluated under the identical protocol and remains below the much larger pretrained TOPIQ-NR model. Verbatim values: `results/koniq_eaqi_results.json`, `results/koniq_baselines_seed0_test.json`.

6.4. Metrics, Calibration, and Uncertainty

The executed KonIQ run of Section 6.2 reports SRCC, KRCC, PLCC (after 4PL calibration on the disjoint calibration partition), and RMSE per seed together with the mean and sample SD across the three seeds (Section 6.5); per-seed predictions and fitted calibration parameters are stored verbatim in `results/koniq_eaqi_results.json`. The same statistics are computed for each training-free baseline on the identical test partition. RMSE is reported on the normalized $[0, 1]$ target scale used by the run; multiplying by the stored train-split target range ($88.39 - 3.91$) for seed 0 converts it to KonIQ-10k MOS units (seed-0 E-AQI: $0.124 \times 84.48 \approx 10.5$ MOS points).

6.5. Metric Definitions and Comparison Rules

SRCC and KRCC are computed on polarity-corrected raw predictions; PLCC and RMSE are computed only after fitting a monotone calibration on a disjoint calibration partition. One explicit four-parameter logistic is:

$$f(x) = \beta_2 + \frac{\beta_1 - \beta_2}{1 + \exp(-(x - \beta_3)/\beta_4)}. \tag{16}$$

The fit enforces $\beta_4 > 0$ and reports x oriented so that larger values indicate better quality (or both prediction and target are oriented consistently toward severity); the fitted parameters accompany every reported PLCC/RMSE pair. Learned configurations use at least three documented seeds, and seed-level predictions are retained; per-seed metrics are summarized by their mean and sample SD, and seed variation is reported separately from test-image uncertainty. Model comparisons on the same images use paired, source-grouped bootstrap resampling with 2,000 resamples applied to both methods under a prespecified stratification rule, with the multiplicity handling and comparison family reported.

Table 6: Component-wise ablation on the fixed seed-0 KonIQ-10k test partition (2,015 images; identical protocol to Table 5, executed with the released `ablate_koniq.py`). All variants share the identical reduced 3-epoch budget and seed. SRCC/KRCC on raw predictions; PLCC/RMSE after 4PL calibration on the disjoint calibration partition. Numeric cells are verbatim outputs stored in `results/koniq_ablation.json`.

Variant	SRCC	KRCC	PLCC	RMSE
(a) full E-AQI	0.663	0.475	0.705	0.131
(b) w/o clipping branch C	0.665	0.477	0.709	0.130
(c) w/o edge branch E	0.663	0.475	0.703	0.131
(d) w/o context branch Z	0.670	0.480	0.706	0.131
(e) hard binary saturation	0.662	0.474	0.704	0.131
(f) fixed cutoffs (5/250)	0.661	0.473	0.703	0.131
(g) concat fusion (no attention)	0.650	0.463	0.682	0.135
(h) MSE only (λ_r=0)	0.657	0.468	0.712	0.129

6.6. Cross-Dataset Transfer

For target dataset T, define the in-domain score using a model trained on T's training partition and tested on T's held-out test partition. Compare it with a source-trained model tested on the identical T test partition without target fine-tuning:

$$\Delta_{\text{gen}}(T) = \rho_{\text{in}}(T) - \rho_{\text{source}\to T}(T). \tag{17}$$

Use matching architecture, target polarity, test images, and seed reporting. A negative gap is possible and is preserved; it is not evidence of an error by itself. Every prespecified target is reported, and an unweighted mean over the stated target set is computed without post-hoc exclusions.

6.7. Executed Ablation Study

An executed component-wise ablation isolates the contribution of each branch on the fixed seed-0 KonIQ-10k split of Section 6.2, with hyperparameters selected on the validation partition and the test partition used only for final reporting (Table 6). Variant (a) is the full prototype. Variants (b)–(d) remove one branch at a time by zeroing its lattice channels and skipping its projection, (e) replaces the soft two-sided saturation response with hard binary thresholding at the same cutoffs, (f) freezes the adaptive cutoffs at their default values (5/250), (g) replaces cross-attention fusion with channel concatenation plus a 1×1 convolution at the common lattice, and (h) removes the pairwise ranking term ($\lambda_r = 0$). Binary versus soft saturation is compared at identical cutoffs so that threshold adaptation is isolated in (f).

| Three findings follow from Table 6. First, under the equal 3-epoch budget, single-branch removals (b)–(d) change SRCC by

at most +0.007 relative to the full prototype: the branches are largely redundant for *general-quality* prediction on KonIQ-10k, which is expected because their individual contributions target the clipping-specific tasks probed by Section 5.1 rather than overall quality ranking; the location-sensitivity result of Section 5.1, not this table, is the evidence that the branch structure earns its place. Second, the fusion mechanism matters beyond any single branch: replacing cross-attention with concat-plus-bias fusion (g) costs 0.013 SRCC and 0.024 PLCC, the largest drop in the table, confirming that the location-conditioned attention is the component that converts map statistics into useful context. Third, the saturation-response and loss variants are within noise of the full model at this budget ((e) and (f) differ by ≤ 0.001–0.002 SRCC; (h) trades a small SRCC loss for a better PLCC/RMSE), so the ranking-term and soft-response choices are justified by the location study rather than by overall-quality correlation. Because the budget is reduced, the absolute numbers are not comparable to Table 5; only within-table differences are interpretable, and the honest reading is that the ablation rules out harm from the added branches on general quality while the clipping-specific benefit is established by the location and human studies.

6.8. Efficiency and Failure Diagnosis

Profiling reports total and trainable parameters separately, including frozen modules, and states input resolution, preprocessing scope, MAC/FLOP convention, precision, batch size, device, software stack, synchronization, warm-up, and repetitions; latency distributions and measured peak memory are reported alongside. Weight storage is a separate quantity from inference memory, and desktop FP16 timing and mobile INT8 timing are never combined into a single latency claim.

Failure cases are inspected using stable image IDs, signed error, and branch interventions. Noise-induced false edges could lower a boundary-conditioned penalty; saturation introduced by clipping noisy values, entropy inflation, and structural-weight suppression are alternative hypotheses requiring interventions. Pre-denoising with an identified NAFNet [33] configuration trained on an appropriate resource such as SIDD [34] is a prospective mitigation with no measured recovery or overhead in this work.

7. Limitations and Release Requirements

E-AQI is a proposed representation and protocol with synthetic operator checks, two computational studies, and an executed 51-observer human study. The framework has not established improved quality correlation, calibrated saturation probability, robustness, compact-model feasibility, or HDR validity. The numerical operator is a heuristic whose thresholds and adaptation ratio may be sensitive to content. Multi-resolution attention may be expensive at high spatial resolutions. Freezing a saliency model does not establish domain suitability or remove its inference cost.

The computational studies carry their own limits. The clipping stimuli are gain-based and bright-endpoint only; shadow clipping, mixed polarity, and learned enhancement outputs are not yet covered. The Sobel–Otsu edge mask is a fixed detector choice whose sensitivity should be re-tested with a learned boundary detector. Base photographs come from BSDS500 rather than from the low-light domains the framework ultimately targets. The IQM verdicts inherit the training distributions of their models (TOPIQ-NR from KonIQ-10k), so their location sensitivity may reflect general-quality statistics rather than clipping-specific perception. The executed human study addresses the perceptual question directly but carries its own scope limits: observers rated on self-reported uncalibrated displays in a single supervised setting, the 2AFC secondary task (H3) was not collected, the within-observer repeat subset was not exercised, and the stimulus pool remains gain-based bright-endpoint clipping on BSDS500 photographs, so shadow-clipping and learned-enhancement stimuli remain untested perceptually. Latency figures are single-device desktop-GPU FP32 measurements and do not represent mobile deployment.

The release accompanying this paper provides the reference implementation (operators, objective, metrics, statistics, profiling harness, the synthetic-check and study scripts that produce every value above), the fixed 70/10/20 split manifests, the trained seed checkpoints, and the complete human-study platform (stimulus generator, server, observer interface, admin dashboard, and analysis script). The anonymized raw ratings, the observer demographics, and the outcome tables of the executed human study are released under the same manifest, together with the platform itself so that the study can be replicated end to end.

The next research steps are testing the location-conditioned assessor against the measured human effect (including a model-human correlation analysis on the released per-stimulus ratings), collecting the 2AFC and repeat extensions on independent observer pools, and independent low-light benchmark evaluation. Enhancement training with the smooth surrogate and extension to video are subsequent studies.

8. Conclusion

We provide a consistent two-sided saturation operator with a fully specified 0.43M-parameter prototype, an executed component-wise ablation, two executed computational studies, and an executed 51-observer human study for clipping assessment. Synthetic numerical checks confirm endpoint behavior and bit-depth code-value equivalence. The computational studies quantify the measurement gap that motivates the framework: on 168 area-matched pairs, three classical no-reference measures are insensitive to clipping location while a deep measure detects it consistently, and across 480 enhancement levels no measure couples strongly to clipping dose. The human study supports the perceptual reading of that gap: at matched clipped area, interior-texture clipping is rated more annoying than edge-aligned clipping (+5.3 on 0–100, $d_z = 0.89$, $p = 3.4 \times 10^{-7}$, 82% of observers), with all five conditions separated (Friedman $p = 2.1 \times 10^{-40}$) and a positive pair-level effect in two-thirds of the image contents—evidence that clipping location matters perceptually under controlled conditions, not evidence

that E-AQI is a validated general-purpose quality index. An executed quality-transfer run on KonIQ-10k (SRCC 0.708±0.011, PLCC 0.751 ± 0.012 over three seeds, above all identically evaluated non-retrained baselines and below the much larger pretrained TOPIQ-NR) quantifies what the from-scratch prototype learns without external pretraining. The evaluation protocol distinguishes annoyance from overall quality, participant effects from stimulus effects, and target-specific generalization from comparisons across unrelated datasets. The released platform, anonymized ratings, and analysis scripts make the perceptual result directly replicable, and the remaining extensions—shadow and learned-enhancement clipping, the 2AFC task, and model–human correlation on per-stimulus ratings—are specified on the same preregistered protocol.

Declaration of generative AI in the writing process

During the preparation of this work the authors used an AI-assisted writing tool in order to improve readability and language consistency. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data and code availability

The reference implementation (operators, objective, metrics, statistics, profiling harness, the synthetic-check scripts that produce every value in Table 1, and the two computational-study scripts that produce every value in Tables 2 and the dose-response section) accompanies this paper. The clipping-location and dose-response studies use the publicly available BSDS500 photographs and the IQA-PyTorch reference implementations of BRISQUE, NIQE, PIQE, and TOPIQ-NR with their published checkpoints; per-stimulus scores are stored verbatim in `results/clipping_location_study.json` and `results/enhancement_clip_response.json`, and the GPU profile in `results/latency_gpu.json`. The KonIQ-10k transfer run is fully reproducible with the accompanying training and evaluation scripts; seed-level predictions, per-seed metrics, and fitted calibration parameters are stored `results/koniq_eaqi_results.json` and `results/koniq_baselines_seed0_test.json`. The complete human-study platform (stimulus generator `make_stimuli.py`, server `server.py`, observer interfaces, admin dashboard, and analysis script `analysis.py`) is included in the release, together with the anonymized raw ratings of the executed 51-observer study (32,130 trials, one row per rating), the anonymized observer demographics, the stimulus manifest with per-stimulus matching metadata, and the analysis outputs (`results/human_study_analysis.json`); no directly identifying observer information was collected.

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